



Final Project
Expository Writing
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Debunking the Myth of a Flat Earth

Since ancient times, humans have believed that the Earth was flat like a table, rather than round like a marble. This belief persisted for hundreds of years until new evidence came to light. However, for the past few hundred years, humans have widely known that the Earth is round. These ideas are widely corroborated by the circumnavigation of various explorers, the laws of physics and gravity, the amazing feat of space exploration perpetrated by humans in the previous century, and, last but not least, the ironclad process of peer reviewed scientific study.

Humans have known that the Earth was round since ancient times. Greek scholars like Pythagoras and Aristotle deduced the spherical shape of the planet before the birth of Christ, however back then, this idea was not widely known. The idea that the Earth was a sphere was given by Eratosthenes, as he was the first person to calculate Earth's circumference (Britannica). But it was proved practically by the voyages of the Portuguese explorer Magellan, who discovered a variety of maritime trade. "The expedition was led by Portuguese explorer Ferdinand Magellan and Spanish navigator Juan Sebastián Elcano, and it set sail from Spain in 1519 with the goal of reaching the Spice Islands of Indonesia. After a perilous journey that lasted over three ..., the expedition finally arrived in the Philippines in 1522. The surviving members of the expedition then continued to China, becoming the first Europeans to circumnavigate the globe" (Winegard). Observations of changing star positions, the curvature of the horizon, and the systematic navigation across vast oceans all pointed to Earth's spherical shape. Magellan thus informed the people of Europe of his discovery.

The roundness of the Earth was further proved during the last century, when the space programs of the Cold War discovered amazing new information about the Earth and other planets in the solar system. The Americans and the Soviets, in their race to prove themselves technologically superior, provided humanity with priceless knowledge on the inner workings of the Earth (National Geographic). Astronauts and scientists aboard spacecraft have testified to the curvature of the Earth's surface. As if this was not enough to convince the skeptics, they also provided various high-resolution snapshots being taken of Earth, including the iconic "Blue Marble" image taken by NASA scientists.

The spherical nature of the Earth has been proved by a body of experts in various fields and is supported by an immense body of evidence. Specialists in diverse fields such as physics, astronomy and geology have all corroborated the spherical nature of our home planet through countless research papers and articles published in the past century. Mathematical models, gravitational principles, and empirical data converge to reaffirm the planet's curvature, reinforcing the scientific consensus and shattering old myths and perceptions that were rooted in humanity for millennia past. NASA Space Place says that “A planet's gravity pulls equally from all sides. Gravity pulls from the center to the edges like the spokes of a bicycle wheel. This makes the overall shape of a planet a sphere” (NASA Space Place, 2024). Furthermore, astrological events like eclipses provide proof for round shape of Earth such that, “During a lunar eclipse the shadow of the Earth always had a round profile...The only object that casts a circular shadow no matter how you shine a light across it is a sphere.” (Steffen, Bruzda, 2019). These incredible collaborations by countless authorities have proved that humanity’s thirst for knowledge is truly unquenchable, and that all is inconsequential when faced with the indomitable human spirit.

Despite the massive amounts of evidence, some people doubt the idea that the Earth is round. So called “flat-earthers” harbor ridiculous ideas of the world lying to them, and them being the only people smart enough to see through the deceit. A popular argument cited by flat-earthers is the argument that the horizon is not curved when they observe it through the naked eye. This argument, however, fails to accommodate the behavior of the horizon when the observations are taken over vast distances, as were taken by Magellan and his contemporaries. The subtle curve of the horizon, the observation that far-flung objects appear bottom first, and the geometric calculations of long-distance observations all align with the spherical Earth model, hence definitively proving the idea that the world is, indeed, round.

In conclusion, the findings of various scientists and explorers over the centuries that make up the human era, the priceless knowledge given to humanity by modern technology and its advancements, and of course the staggering amount of evidence that discredits the flat-earth theory, all are united in pointing to the irrefutable fact that the world is as round as a perfect ball. From the intrepid circumnavigation of Magellan to the terrifying, yet beautiful, modern-day space missions, humanity's journey of discovery has unveiled the planet's true form—a majestic blue sphere suspended effortlessly in the inescapable abyss of

space. As humanity grows and learns more, we embrace the scientific method and slowly discard the archaic beliefs of the ancients. The undeniable truth of the curvature of Mother Earth stands as a testament to the herculean efforts of the scientists and professionals, as well as to the vastness of human curiosity and the adamant human spirit.

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